ML Preprocessing Cheat Sheet

Before training any ML model, ALWAYS follow this pipeline:
1. Understand Data
2. Clean Data
3. Explore Data (EDA)
4. Create Features
5. Prepare for Model

Rule:

Better data preparation > better model performance

1. Understand the Dataset

Know what you are working with.

Key Questions:

- What does each row represent? (Match / Player / Ball)
- What are the columns?
- What is the target variable? (Winner)

Basic Commands (Pandas)

PYTHON

```
df.head()
df.info()
df.describe()
df.columns
df.shape
```

Things to Notice:

- Data types (int, float, object)
- Missing values
- Irrelevant columns

2. Data Cleaning

Make data usable and consistent

2.1 Handle Missing Values

PYTHON

```
df.isnull().sum()
```

Options:

Remove rows:

PYTHON

```
df.dropna()
```

Fill values:

PYTHON

```
df.fillna(0)
df.fillna(df.mean())
```

Rule:

- Small missing → fill
- Large missing → consider dropping column

2.2 Fix Inconsistent Data

Example:
"MI", "Mumbai Indians", "mumbai indians"

PYTHON

```
df['team'] = df['team'].str.lower()
```

Standardize naming

2.3 Remove Duplicates

PYTHON

```
df.drop_duplicates()
```

2.4 Correct Data Types

PYTHON

```
df['date'] = pd.to_datetime(df['date'])
df['runs'] = df['runs'].astype(int)
```

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3. Exploratory Data Analysis (EDA)

Find patterns in data BEFORE ML

3.1 Univariate Analysis

Analyze one column at a time

PYTHON

```
df['team'].value_counts()
df['runs'].hist()
```

3.2 Bivariate Analysis

Relationship between 2 variables

PYTHON

```
sns.countplot(x='toss_winner', hue='match_winner', data=df)
```

3.3 Correlation Analysis

PYTHON

```
df.corr()
sns.heatmap(df.corr(), annot=True)
```

Helps find:

- Important features
- Redundant features

3.4 Key Questions for IPL Dataset

- Does toss affect match outcome?
- Does venue affect winning?
- Which team has highest win rate?
- Does chasing vs batting first matter?

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4. Feature Engineering (MOST IMPORTANT)

Create meaningful inputs for model

4.1 Create New Features

Examples:

Win Percentage

PYTHON

```
win_rate = wins / total_matches
```

Recent Performance

- Last 5 matches win %

Head-to-Head Stats

- Team A vs Team B past wins

Venue Advantage

- Team performance at specific stadium

Golden Rule:

"Model learns only what you give it"

4.2 Encoding Categorical Data

ML models need numbers

Label Encoding

PYTHON

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['team'] = le.fit_transform(df['team'])
```

Team	Encoded
MI	0
CSK	1
RCB	2

One-Hot Encoding

PYTHON

```
pd.get_dummies(df, columns=['team'])
```

Team	MI	CSK	RCB
MI	1	0	0
CSK	0	1	0

When to use:

- Few categories → One-hot
- Many categories → Label encoding

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5. Handling Imbalanced Data

Problem:

One class dominates

Example:

- Some teams win way more

Solutions:

- Undersampling
- Oversampling

6. Feature Selection

Remove useless features

Remove:

- ID columns
- Constant columns
- Highly correlated columns

Use:

PYTHON

```
df.corr()
```

Rule:

"Less but relevant features > more noisy features"

7. Train-Test Split

Test model on unseen data

PYTHON

```
from sklearn.model_selection import train_test_split

X = df.drop('winner', axis=1)
y = df['winner']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

8. Feature Scaling (if needed)

PYTHON

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Final Pipeline

```
Load Data
    ↓
Understand Data
    ↓
Clean Data
    ↓
EDA (Find Patterns)
    ↓
Feature Engineering
    ↓
Encoding
    ↓
Feature Selection
    ↓
Train-Test Split
    ↓
{Then Model Training}
```

Collapse

Note:

For more detailed understanding, refer Python for DA course

Find the AIPL GitHub
GitHub